

Yishu Li

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EDUCATION

Brown University

Providence, RI | Expected May 2026

Sc.M. in Biomedical Engineering, 4.0/4.0 GPA

Thesis Topic: Long-term Stable Brain-Computer Interface for Function Restoration

University of California, Irvine

Irvine, CA | Sept 2022 – June 2023

Exchange Student, 3.81/4.00 GPA

Southern University of Science and Technology

Shenzhen, Guangdong, China | Sept 2019 – June 2023

B.Eng. in Biomedical Engineering, 3.64/4.0 GPA

RESEARCH EXPERIENCE

[Google Scholar](#)

BrainGate Team

Brown University | Providence, RI | Sept 2024 – Present

Graduate Student Researcher, under Dr. John Simeral & Dr. Leigh Hochberg

- **Communication Restoration for an Advanced ALS Patient**
 - Designed and implemented iBCI decoders for restoring communication in stroke and ALS patients, utilizing Machine Learning and Deep Learning techniques. The system has been the only communication method for more than 6 months.
 - Enabled communication for an advanced ALS patient with a multiclass fast-deploy Gaussian Naïve Bayes iBCI decoder, reaching over 90% accuracy for 5-way online decoding.
 - Developed a sleep detection method for locked-in ALS patients based on the LFP signal, reaching over 90% accuracy.
- **Meta-Learning for Stable Long-term iBCI Decoder** [Presentation](#)
 - Improve decoding stability by minimizing the need for recalibration and enabling same-day adaptation, reducing the recalibration time to within 1 minute.
 - Train and evaluate the model on the supercomputer OSCAR for higher running speed.

Center for BioMedical Signal Processing and Computation

UC Irvine | Irvine, CA | Oct 2022 – June 2023

Undergraduate Researcher, under Dr. Zoran Nenadic

- **A Study of Muscle Fatigue on Corticomuscular Coherence**
 - Identified a reduction in the μ band of EEG power and corticomuscular coherence, establishing measures as biomarkers for detecting muscle fatigue and enhancing stroke rehabilitation outcomes.
 - Conducted experiment design, data collection, and processing, and authored a thesis for the bachelor's degree.

SUSTech Brain-Robot Rehabilitation Technology Lab

SUSTech | Shenzhen, Guangdong, China | Sept 2020 – June 2022

Undergraduate Researcher, under Dr. Mingming Zhang

- **Characterization of Bimanual Cyclical Tasks from Single-Trial EEG-fNIRS Measurements** [Link](#)
 - Developed a technique to characterize neural activities during robot-assisted bimanual training, achieving 90% classification accuracy in bimanual cyclical tasks with simultaneous EEG and fNIRS measurements.
 - Engaged in neural data recording experiments and data processing using Python and MATLAB.
- **Short-interval Priming Effect: an EEG study of Action Observation on Motor Imagery** [Link](#)
 - Characterized the priming effect of short-duration tasks in rehabilitation, demonstrating flexibility and timely feedback to enhance the application of priming in rehabilitation processes.
 - Demonstrated a more intense neural response and higher classification accuracy induced by the priming effect compared to the control group.
 - Contributed to the entire project, encompassing experiment design, setup, execution, data analysis, and paper writing.
- **Subject-customized fNIRS Probe Hat Design**
 - Developed a pipeline to 3D print subject-customized fNIRS probe hats from MRI scans using FreeSurfer, MATLAB, and SolidWorks, facilitating faster experimental setups and precise probe localization for diverse subjects.

PUBLICATIONS

- Y. -C. Jiang et al., "Characterization of Bimanual Cyclical Tasks From Single-Trial EEG-fNIRS Measurements," in IEEE Transactions on Neural Systems and Rehabilitation Engineering, vol. 30, pp. 146-156, 2022, doi: 10.1109/TNSRE.2022.3144216. [Read Here](#)
- Z. Sun, Y. -c. Jiang, Y. Li, J. Song and M. Zhang, "Short-term Priming Effects: an EEG Study of Action Observation on Motor Imagery," in IEEE Transactions on Cognitive and Developmental Systems, 2022, doi: 10.1109/TCDS.2022.3187777. [Read Here](#)

INTERNSHIP EXPERIENCE

Shanghai Institute of Microsystem and Information Technology, Chinese Academy of Sciences

Algorithm Intern, Dr. Meng Li's Group

Shanghai, China | July 2023 – Apr 2024

- **Brain Connectivity Changes in Different Visual States**

- Investigated brain connectivity changes during visual search and recognition processes to better understand the top-down and bottom-up visual pathways and the mechanism of attention.
- Led the entire project, including building the virtual reality experiment in Unity, conducting clinical experiments, programming the sEEG data analysis pipeline, and performing spike sorting and analysis.

Shenzhen 2nd People's Hospital, Neurosurgery Department

Shenzhen, Guangdong, China | July 2022 – Aug 2022

Summer Associate, Technician Intern

- Conducted sEEG recording experiments on patients with epilepsy to advance research and improve patient outcomes.
- Assisted in preparing TMS sessions for patients with depression or pain, ensuring efficient setup and patient comfort.

COURSE PROJECTS

- **NeurANIL: A Meta-learning framework for brain-to-text decoding | Deep Learning** [Poster](#) Brown | Spring 2025

- Designed and implemented a meta-learning algorithm based on the 'Almost No Inner Loop' (ANIL) framework for brain-to-text phoneme decoding.

- **PCA-GNB Model for Classification | Machine Learning: From Theory to Algorithm** [Report](#) Brown | Fall 2024

- Enhanced Gaussian Naïve Bayes performance by applying Principal Component Analysis to eliminate feature dependencies, resulting in higher performance on all scikit-learn built-in classification datasets.

- **Sudoku GUI | Java Programming** SUSTech | Spring 2020

- Developed a Sudoku game with a GUI using Java, featuring complete functions such as data generation, hints, and error correction.

- **Throwing Robot Modeling | Engineering Drawing** SUSTech | Summer 2020

- Designed and modeled a throwing robot using SolidWorks, incorporating two motor generators, a Raspberry Pi board, and two Mecanum wheels.
- Developed a video showcasing the robot, utilizing SolidWorks' visual rendering functions for enhanced presentation.

- **EEG Data Generation Using ResNet | Machine Learning and Neuro Engineering** SUSTech | Spring 2021

- Enhanced 32-channel EEG signal data into 64-channel outputs using a ResNet architecture.
- Preprocessed and visualized data to enhance analysis and interpretation using the Python Package MNE.

- **Gesture Identification based on OpenCV | Medical Image Processing** SUSTech | Fall 2021

- Developed a gesture-based program using feature extraction and Convex Hull Detection with Python OpenCV to control the computer cursor, enabling mouse movement and clicks based on user gestures.

EXTRACURRICULAR ACTIVITIES

SUSTech Men's Soccer Club

Shenzhen, Guangdong, China | Sept 2020 – June 2022

Chair

- Coordinated and executed 3 university-level soccer competitions, enhancing student engagement and athletic participation.

SKILLS

Programming: Python, MATLAB, Java, C#, Linux, PyTorch, Tensorflow, Scikit-learn, Git, Wandb.

Software: Microsoft Office, AutoCAD, SolidWorks, Unity, Offline Sorter, Neuroexplorer, Freesurfer.

Biosignal Processing: EEG, MRI, fNIRS, Spike train, Local field potential.

Analytic Skills: Signal processing, Machine learning, Deep learning, Temporal data analysis, Network analysis, Data visualization, 3D modeling and printing, Clinical communication, Clinical experiment design and conduction.

Language: Mandarin (Native), English (Fluent), Spanish (Elementary).

HONORS & AWARDS

2019 – SUSTech Freshman Scholarship, First Prize (¥30,000.00)

2024 – Brown University Institute of Biology, Engineering, and Medicine Master's Scholarship (\$8,567.00)

2025 – Brown University Academic Sub-track Summer Research Stipend (\$5,000.00)